

MINNEAPOLIS-ST PAUL, MN, USA

WMO: 726580

Lat: 44.883N

Lon: 93.229W

Elev: 266

StdP: 98.17

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14922

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.6	-21.1	-28.3	0.3	-23.0	-25.9	0.4	-20.3	11.3	-9.0	10.2	-8.7	3.6	310	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	32.7	23.0	31.1	22.2	29.5	21.3	25.0	30.7	23.8	29.0	22.8	27.8	5.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.5	28.6	22.1	17.3	27.4	21.0	16.1	26.1	77.4	30.6	72.6	29.1	68.4	27.9	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.6	9.2	8.3	-27.1	35.7	3.4	1.7	-29.6	36.9	-31.6	37.9	-33.5	38.8	-36.0	40.0		
			-27.3	26.8	3.2	1.1	-29.7	27.6	-31.5	28.3	-33.4	28.9	-35.7	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-9.0	-6.5	0.6	8.4	15.2	20.9	23.6	22.1	17.8	10.0	1.8	-5.7	(6)
(7)		DBStd	12.55	7.08	6.75	6.76	5.60	4.79	3.80	3.17	3.03	4.63	5.12	5.92	6.42	(7)
(8)		HDD10.0	2258	588	463	299	94	10	0	0	2	64	252	486		(8)
(9)		HDD18.3	4109	846	697	550	301	122	18	2	64	264	496	744		(9)
(10)		CDD10.0	1654	0	0	8	46	171	326	422	376	237	63	6	0	(10)
(11)		CDD18.3	463	0	0	1	2	24	94	166	122	49	5	0	0	(11)
(12)		CDH23.3	3927	0	0	2	27	241	794	1513	926	383	41	0	0	(12)
(13)		CDH26.7	1256	0	0	0	6	76	253	543	264	106	7	0	0	(13)
(14)	Wind	WSAvg	4.1	4.1	4.2	4.4	4.8	4.5	4.0	3.7	3.5	3.9	4.3	4.2	4.0	(14)
(15)	Precipitation	PrecAvg	738	25	20	47	62	90	107	95	74	57	42	24		(15)
(16)		PrecMax	1015	92	54	121	178	204	249	455	236	175	144	134	108	(16)
(17)		PrecMin	419	2	2	8	4	16	6	15	19	12	5	2	5	(17)
(18)		PrecStd	155	19	14	26	36	45	55	74	49	39	41	35	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	11.3	21.4	26.9	32.4	34.0	35.5	33.6	32.0	27.8	19.6	10.3	(19)
(20)		MCWB	3.5	7.7	14.5	16.5	20.1	23.6	24.8	23.6	21.8	18.0	12.9	7.7		(20)
(21)		2%	DB	3.8	7.1	16.8	22.5	28.6	31.6	33.0	31.2	29.5	23.8	15.7	6.7	(21)
(22)		MCWB	1.8	4.4	11.8	13.5	18.0	21.7	23.9	22.7	21.3	16.3	11.0	4.2		(22)
(23)		5%	DB	1.9	4.6	13.3	20.0	26.0	29.5	31.3	29.4	27.5	20.7	12.7	4.3	(23)
(24)		MCWB	0.5	2.3	8.9	11.7	17.0	20.7	22.8	21.8	20.0	14.6	8.5	2.3		(24)
(25)		10%	DB	0.4	2.6	9.9	17.4	23.3	27.7	29.7	27.9	25.4	18.0	10.1	2.4	(25)
(26)		MCWB	-0.8	0.7	6.3	10.3	15.4	19.5	21.9	20.9	18.9	13.1	6.5	0.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	7.8	15.9	17.4	22.3	25.4	27.0	25.8	24.0	20.7	14.6	8.7	(27)
(28)		MCDWB	6.4	11.2	19.7	24.5	30.2	31.8	32.1	31.4	29.3	24.6	18.0	10.4		(28)
(29)		2%	WB	2.0	4.7	12.3	14.7	20.0	23.4	25.4	24.2	22.6	17.6	11.6	4.3	(29)
(30)		MCDWB	3.5	7.1	16.3	20.8	25.5	29.3	31.3	28.8	27.4	21.4	15.2	6.2		(30)
(31)		5%	WB	0.7	2.4	9.3	12.8	18.3	22.0	24.2	23.2	21.3	15.4	8.9	2.4	(31)
(32)		MCDWB	1.9	4.4	13.0	18.5	23.8	27.6	29.6	27.8	25.7	19.6	12.3	4.0		(32)
(33)		10%	WB	-0.7	0.8	6.4	11.0	16.7	20.7	23.0	22.0	20.0	13.5	6.8	1.0	(33)
(34)		MCDWB	0.4	2.4	9.8	16.6	22.2	25.9	28.0	26.4	24.3	17.4	9.7	2.3		(34)
(35)	Mean Daily Temperature Range	MDBR	7.9	8.3	8.8	10.0	10.2	9.7	9.6	9.4	9.9	8.9	7.6	7.0		(35)
(36)		5% DB	MCDBR	8.3	9.3	12.7	14.0	13.7	11.8	11.0	10.6	11.7	12.9	11.3	9.3	(36)
(37)		MCWBR	6.9	7.1	8.1	7.7	6.3	5.3	4.8	4.8	5.5	7.3	7.4	7.3		(37)
(38)		5% WB	MCDBR	8.1	8.9	11.8	12.6	12.1	10.3	10.2	9.6	10.3	10.9	10.5	8.7	(38)
(39)		MCWBR	7.0	7.0	8.0	7.8	6.6	5.7	5.4	5.4	5.9	7.2	7.5	7.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.271	0.286	0.323	0.379	0.408	0.402	0.402	0.406	0.379	0.324	0.294	0.270		(40)
(41)		taud	2.416	2.356	2.379	2.287	2.258	2.323	2.342	2.332	2.372	2.520	2.509	2.437		(41)
(42)		Ebn at Noon	857	904	912	881	862	867	863	845	840	853	825	819		(42)
(43)		Edh at Noon	72	92	104	125	132	125	121	118	105	78	66	64		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.73	2.81	3.78	4.59	5.33	5.93	6.29	5.26	4.10	2.61	1.66	1.21		(44)
(45)		RadStd	0.20	0.29	0.35	0.47	0.43	0.47	0.23	0.38	0.29	0.35	0.16	0.13		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103	+55	(46)
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page